

Shantanu Ghosh^(he/him)

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https://shantanu-ai.github.io/ • 📄 Google scholar
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Research Interests

Method: Robustness, Generalization, Trustworthy Machine Learning, Multimodal learning, Explainable AI.
Applications: Joint modeling of medical images (chest-X-rays or 2D mammograms) and reports for lung diseases (e.g., pneumonia) and breast cancer predictions.

Education

Boston University

Doctor of Philosophy, Electrical Engineering

Thesis: **Interpretable Medical AI with Vision-Language Alignment**

Advisor(s): [Dr. Kayhan Batmanghelich](#)

Committee: [Dr. Brian Kulis](#), [Dr. Wenchao Li](#), [Dr. Clare Poynton](#)

Boston, Massachusetts, USA

Jan 2023 – Dec 2026 (Expected)

University of Pittsburgh (Transferred to BU)

Doctor of Philosophy, Intelligent Systems

Advisor(s): [Dr. Kayhan Batmanghelich](#)

Pittsburgh, Pennsylvania, USA

Aug 2021 – Dec 2022

University of Florida

Master of Science, Computer Science, 3.88/4.00

Advisor: [Dr. Mattia Proserpio](#)

Gainesville, Florida, USA

Aug 2019 – May 2021

Institute of Engineering and Management

Bachelor of Technology, Computer Science, 3.6/4.00

Kolkata, West Bengal, India

Aug 2008 – May 2012

Publications

Preprints

- [P1] **Mammo-FM: Breast-specific foundational model for Integrated Mammographic Diagnosis, Prognosis, and Reporting**
Shantanu Ghosh, Vedant Parthesh Joshi, Rayan Syed, Aya Kassem, Abhishek Varshney, Payel Basak, Judy Wawira Gichoya, Hari M. Trivedi, Imon Banerjee, Clare B. Poynton, Shyam Visweswaran, Kayhan Batmanghelich [\[Paper\]](#)

Conference Proceedings

- [C9] **PhyDiCT: Plug-and-Play CT Reconstruction from Sparse X-Rays via Differentiable Rendering and Strong Priors**
Weicheng Dai, **Shantanu Ghosh**, Kayhan Batmanghelich
International Conference on Medical Image Computing and Computer Assisted Intervention (**MICCAI**), 2026.
- [C8] **LADDER: Language-Driven Slice Discovery and Error Rectification in Vision Classifiers**
Shantanu Ghosh, Rayan Syed, Chenyu Wang, Vaibhav Choudhary, Binxu Li, Clare B. Poynton, Shyam Visweswaran, Kayhan Batmanghelich
Findings of 2025 Conference on Association for Computational Linguistics (**ACL**) [\[Project\]](#) [\[Paper\]](#) [\[Code\]](#)
- [C7] **Semantic Consistency-Based Uncertainty Quantification for Factuality in Radiology Report Generation**
Chenyu Wang, Weichao Zhou, **Shantanu Ghosh**, Kayhan Batmanghelich, Wenchao Li
Findings of the Association for Computational Linguistics: NAACL 2025 (**NAACL**). [\[Paper\]](#) [\[Code\]](#)
- [C6] **Mammo-CLIP: A Vision Language Foundation Model to Enhance Data Efficiency and Robustness**

in Mammography

Shantanu Ghosh, Clare B. Poynton, Shyam Visweswaran, Kayhan Batmanghelich
International Conference on Medical Image Computing and Computer Assisted Intervention (**MICCAI**), 2024.
(Early accept, top 11%) [[Project](#)] [[Paper](#)] [[Reviews](#)] [[Code](#)]

- [C5] ***Distilling BlackBox to Interpretable models for Efficient Transfer Learning***
Shantanu Ghosh, Ke Yu, Kayhan Batmanghelich
International Conference on Medical Image Computing and Computer Assisted Intervention (**MICCAI**), 2023.
(Early accept, top 14%) [[Project](#)] [[Paper](#)] [[Reviews](#)] [[Code](#)]
- [C4] ***Dividing and Conquering a BlackBox to a Mixture of Interpretable Models: Route, Interpret, Repeat***
Shantanu Ghosh, Ke Yu, Forough Arabshahi, Kayhan Batmanghelich
International Conference on Machine Learning (**ICML**), 2023. [[Project](#)] [[Paper](#)] [[Code](#)]
- [C3] ***DR-VIDAL - Doubly Robust Variational Information-theoretic Deep Adversarial Learning for Counterfactual Prediction and Treatment Effect Estimation***
Shantanu Ghosh, Zheng Feng, Jiang Bian, Kevin Butler, Mattia Proserpi
American Medical Informatics Association (**AMIA**) Symposium, 2022 (Oral). [[Paper](#)] [[Code](#)]
- [C2] ***Anatomy-Guided Weakly-Supervised Abnormality Localization in Chest X-rays***
Ke Yu, **Shantanu Ghosh**, Zhexiong Liu, Christopher Deible, Kayhan Batmanghelich
International Conference on Medical Image Computing and Computer Assisted Intervention (**MICCAI**), 2022.
[[Paper](#)] [[Code](#)]
- [C1] ***Causal AI with Real World Data: Do Statins Protect From Alzheimer's Disease Onset?***
Mattia Proserpi, **Shantanu Ghosh**, Zhaoyi Chen, Marco Salemi, Tianchen Lyu, Jiang Bian
International Conference on Medical and Health Informatics (**ICMHI**), 2021. [[Paper](#)]

Journal Articles.....

- [J3] ***Anatomy-specific Progression Classification in Chest Radiographs via Weakly-Supervised Learning***
Ke Yu, **Shantanu Ghosh**, Zhexiong Liu, Clare Poynton, Christopher Deible, Kayhan Batmanghelich
Radiology: Artificial Intelligence, (**RAD: AI**), **IF:8.1**, 2024 [[Paper](#)] [[Code](#)].
- [J2] ***Propensity Score Synthetic Augmentation Matching using Generative Adversarial Networks (PSSAM-GAN)***
Shantanu Ghosh, Christina Boucher, Jiang Bian, Mattia Proserpi
Journal of Computer Methods and Programs in Bio-medicine Update, 2021. [[Paper](#)] [[Code](#)]
- [J1] ***Deep Propensity Network using a Sparse Autoencoder for Estimation of Treatment Effects***
Shantanu Ghosh, Jiang Bian, Yi Guo, Mattia Proserpi
Journal of the American Medical Informatics Association (**JAMIA**), **IF:4.7**, 2021. [[Paper](#)] [[Code](#)]

Peer reviewed workshops.....

- [W3] ***Distributionally robust self-supervised learning for tabular data***
Shantanu Ghosh, Tiankang Xie, Mikhail Kuznetsov
Table Representation Learning Workshop (**TRL**), NeurIPS, 2024. [[Paper](#)] [[Code](#)]
- [W2] ***Tackling Shortcut Learning in Deep Neural Networks: An Iterative Approach with Interpretable Models***
Shantanu Ghosh, Ke Yu, Forough Arabshahi, Kayhan Batmanghelich
Workshop on Spurious Correlations, Invariance and Stability (**SCIS**), ICML, 2023. [[Paper](#)] [[Poster](#)]
- [W1] ***Bridging the Gap: From Post Hoc Explanations to Inherently Interpretable Models for Medical Imaging***

Shantanu Ghosh, Ke Yu, Forough Arabshahi, Kayhan Batmanghelich
Workshop on Interpretable Machine Learning in Healthcare (**IMLH**), ICML, 2023. [\[Paper\]](#) [\[Poster\]](#)

Research Experience

Boston University.....

Graduate Research Assistant

Boston, Massachusetts, USA

Advisor(s): [Dr. Kayhan Batmanghelich](#), [Dr. Clare B. Poynton](#)

Jan 2023 – Present

- o Developed **LADDER**, a slice discovery and mitigation algorithm using vision language (VLM) models and LLMs to reason and fix the classifier's mistakes. **Accepted at ACL, 2025 (Findings)**
- o Developed **Mammo-CLIP** the first vision language foundation model for 2D mammograms. **Accepted at MICCAI, 2024 (top 11%)**
- o Applied the mixture of interpretable models for efficient transfer learning to an unseen domain with limited training data. **Accepted at MICCAI, 2023 (top 14%)**.

University of Pittsburgh.....

Graduate Student Researcher

Pittsburgh, Pennsylvania, USA

Advisor(s): [Dr. Kayhan Batmanghelich](#), [Dr. Forough Arabshahi](#)

Aug 2021 – Dec 2022

- o Developed an iterative algorithm to extract a mixture of interpretable models from a Blackbox to provide instance-specific First-order logic-based explanations using human-understandable concepts. **Accepted at ICML, 2023**.

University of Florida.....

Graduate Research Assistant

Gainesville, Florida, USA

Advisor(s): [Dr. Kevin Butler](#), [Dr. Jiang Bian](#), [Dr. Mattia Proserpi](#)

Jan 2020 – Jul 2021

- o Developed novel deep learning frameworks to estimate propensity score, namely DPN-SA (**JAMIA 2021**), PSSAM-GAN (**CMPB-U 2021**), and DR-VIDAL (**AMIA 2022, oral**), for the efficient estimation of individual treatment effects (ITE).

Industry Experience

United Imaging Intelligence (UII-America).....

Research Scientist Intern

Boston, MA, USA

Computer Vision and Robotics. Mentor: [Dr. Ziyang Wu](#)

May 2026 – Aug 2026

- o Developing Video LLM for surgical video understanding.

Amazon.....

Applied Scientist Intern

Pasadena, California, USA

AWS Optimus. Mentor: [Dr. Ankan Bansal](#)

May 2025 – Aug 2025

- o Developed an algorithm to detect and mitigate biases in the multi-modal LLM coding agents.

Amazon.....

Applied Scientist Intern

NYC, New York, USA

AWS, Security Analytics and AI Research (SAAR). Mentor: [Dr. Mikhail Kuznetsov](#)

Jun 2024 – Sep 2024

- o Developed a framework to learn robust representations to fix systematic errors in pre-trained self-supervised models for tabular data. Publication at **TRL@NeurIPS 2024**.

Lexmark International India Pvt Ltd.....

Software Engineering Professional II

Kolkata, India

Oct 2016 – Jul 2019

- o Developed the ISP component of the product **Publishing Platform for Retail (PPR)**.

Cognizant Technology Solutions India Pvt Ltd.....

Associate, Projects

Kolkata, India

Mar 2013 – Sep 2016

- o Developed **WCF** web services in the Contract First Approach using Service Oriented Architecture.

Honors & Awards

- o **Achievement Award** of 4500 USD during the admission of graduate studies at UF in Fall 2019.
- o **Star Employee** award in Q4, 2013 and Q4, 2015 in Cognizant Technology Solutions.
- o **Outstanding reviewer** in NeurIPS 2024, conference registration waived.
- o **Travel Award** for poster presentation at [SAIL 2026 Symposium](#).

Academic Service

Journal Review.....

Transactions on Machine Learning Research (TMLR), IEEE Transactions on Medical Imaging (IEEE-TMI), Journal of Biomedical Informatics (JBI), Medical Image Analysis (MedIA), Journal of the American Medical Informatics Association (JAMIA), Journal of Computer Methods and Programs in Biomedicine (CMPB), Biometrical Journal, Information Fusion

Conference Review.....

BMVC (2026), ECCV (2026), WACV (2025), ICCV (2025), ICML (2025), ICLR (2024-2026), AAAI (2024-2026), AISTATS (2025), NeurIPS (2023-2026), MICCAI (2024-2026), CVPR (2024-2026), CLeaR (2024, 2025)

Teaching Experience

- o Special Topics: Medical Imaging With AI (EC 500) - Guest Lecturer (Fall 2025)
- o Deep Learning (EC 523) - Teaching Assistant (Fall 2024)
- o Introduction to Software Engineering (EC 327) - Teaching Assistant (Fall 2023)

Student Mentoring

- o [Rayan Syed](#), Undergraduate Student, Boston University (**BU outstanding undergrad research award**)
- o [Ho Cheung Aiden Wong](#), Undergraduate Student, Boston University (Next: MS, Columbia University)
- o [Alex Tang](#), Undergraduate Student, Dartmouth University
- o [Ryan Gupta](#), Undergraduate Student, Boston University
- o [Abhishek Varshney](#), Masters Student, Boston University
- o [Akshat Gurbuxani](#), Masters Student, Boston University

Talks & Abstract Presentation

- o *DR-VIDAL for Counterfactual Prediction and Treatment Effect Estimation*, Oral Talk, AMIA 2022 Annual Symposium, Nov 2022 [\[Talk\]](#) [\[Slides\]](#)
- o *Divide and Conquer: Carving Out Concept-based Models out of BlackBox for More Efficient Transfer Learning*
 - Fall ISP AI Forum, University of Pittsburgh, Nov 2023 [\[Talk\]](#) [\[Slides\]](#)
 - MedAI Group, Stanford University, Oct 2024 [\[Talk\]](#) [\[Slides\]](#)
- o *A Domain-Specific Foundation Model Surpassing Generalist AI for Integrated Mammographic Diagnosis, Prognosis and Reporting*
 - Annual Boston Medical Imaging Workshop, MIT CSAIL, Oct 2025 [\[Slides\]](#)
- o *Text-Grounded Interpretability for Mammography Risk Prediction via Vision-Language Alignment*
 - [Symposium on Artificial Intelligence for Learning Health Systems \(SAIL\) 2026](#)
 - [International Breast Density & Cancer Risk Assessment Workshop \(IBDW\) 2025](#)